

Alignment-Preserving Hierarchical Hypothesis Management for Difficult Open-Ended Differential Diagnosis

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July 29, 2026

Abstract

Difficult open-ended differential diagnosis requires recalling enough disease hypotheses and keeping candidate representations and belief states consistent across stages of hierarchical reasoning. Simply enlarging the candidate space creates synonym fragmentation, local evidence that never updates the global decision state, and output names that drift from canonical concepts. We retain the Recall–Discrimination Dilemma as motivation, but upgrade the research object to alignment-preserving hierarchical hypothesis management (APHHM): case-adaptive organization, candidate-relative evidence updating, equivalence-class-aware candidate compression, local posterior write-back, and bounded global decoding. All reported scores use an unmodified scoring interface: we deliberately exclude synonym bind-repair, because that interface-level lever raises baseline scores at least as much as ours and therefore measures protocol alignment rather than reasoning quality. On frozen 100-case slices of DiagnosisArena (DA), MedCaseReasoning (MCR), and Open-XDDx (OX), the hierarchical stack reaches DA option @1/@2 of 0.71/0.78 versus 0.62/0.71 for the strongest external baseline. A case-irrelevant random L1 axis costs 0.34 option @1 on DA and collapses candidate-relative evidence selection (mean selected facts $4.94 \rightarrow 0.13$). State-consistent write-back at locked budget raises OX micro-F1 from 0.576 (cold) to 0.651 (paired bootstrap $p=4.0 \times 10^{-4}$); budget alone moves at most 0.008, and per-family caps are not load-bearing. Pre-registered ablations locate the compression contribution in the structural requirement rather than in our heuristic: removing equivalence handling costs 0.102 accuracy on MCR (about a fifth of the deployed configuration’s), a candidate-count-matched semantics-blind control is significantly worse, yet the gate predicate, its timing and its composition are equivalent within ± 3 pp—and we retract an earlier closed-set option-accuracy gain that an audit shows to be a scoring artifact. A genuinely budget-matched flat control (ten-sample self-consistent retrieve–rerank, roughly 90 model calls per case) does not close these gaps. We insist on stage-attributable errors—TREE_PARENT_ABSENT, L2_MISS, LOCAL_ELIMINATION, GLOBAL_MISRANK, MAPPER_UNBIND—because our audits show that conflating scoring-interface failures with recall failures recommends interventions that measurably reduce accuracy (full-tree leaf injection: option @1 $0.72 \rightarrow 0.42$). Candidate-relative evidence selector strategies (including anti-prototype anchoring) remain open components; what the ablations support is the case-adaptive axis that feeds them, not a particular selection rule.

Terminology at a Glance

LLM: large language model; RAG: retrieval-augmented generation; CoT: chain-of-thought; L1/L2: Level-1 diagnostic family / Level-2 concrete disease leaf; DA/MCR/OX: DiagnosisArena / MedCaseReasoning / Open-XDDx; APHHM: Alignment-Preserving Hierarchical Hypothesis Management (this paper); RRF: reciprocal rank fusion; Fine crowding: multiple near-synonym L2 leaves occupying top frontier slots; soft prior: overturnable L1 relative score used in global

*Draft revised to an alignment-preserving narrative. Not a camera-ready submission.

arbitration; SC: self-consistency (multi-sample decoding with rank fusion); A1/A2/A3: the representation, state, and interface alignment invariants defined in Section 3.2. Failure-stage labels are defined in Section 3.3. A fuller glossary is in Appendix A.

1 Introduction

Large language models (LLMs) can propose differential diagnoses from clinical vignettes, and retrieval-augmented generation (RAG) can enlarge the set of disease hypotheses drawn from case reports, guidelines, and textbooks. Difficult open-ended diagnosis is therefore often framed as a tension between coverage and ranking quality. We still call this tension the Recall–Discrimination Dilemma: increasing recall admits the correct disease more often, but also admits redundant, near-duplicate, and granularity-mismatched hypotheses that make discrimination harder.

That dilemma is necessary but no longer sufficient as a research framing. In a hierarchical pipeline, failures also arise when representations and beliefs fall out of alignment across stages:

- Synonym fragmentation: near-equivalent disease strings compete as separate leaves and consume frontier budget.
- State discontinuity: local evidence updates improve a branch score that is never written back into the tree state used for global decoding.
- Name–concept detachment: the ranked clinical entity is present, yet the evaluation interface fails to bind the output string (or multiple-choice option) to the canonical leaf.

We therefore study alignment-preserving hierarchical hypothesis management (APHHM): case-adaptive organization of hypotheses; candidate-relative evidence updating; equivalence-class-aware candidate compression; local posterior write-back into a shared belief state; and bounded global decoding over that state.

Existing systems typically address only part of this problem. Direct chain-of-thought (CoT) and flat RAG couple retrieval, organization, and ranking in one generation step [7, 5]. Verification methods such as Dual-Inf improve reflection over a flat differential list [11]. Multi-agent frameworks such as MDAgents and MAC diversify roles rather than the geometry of the disease hypothesis space [1, 9]. Fixed specialty or knowledge-graph hierarchies provide reusable routing, not a case-specific comparison coordinate system [2]. Interactive diagnosis benchmarks emphasize information acquisition, which is adjacent to—but distinct from—keeping a large static evidence set organized and state-consistent [8, 6].

Contributions.

1. Case-adaptive hierarchical hypothesis space. We organize multi-source disease recalls into a case-specific, single-axis Level-1 (L1) partition that serves as a temporary comparison coordinate system. Pre-registered ablations show that a case-irrelevant random partition collapses the cascade; gap-fill is retained as a structural repair but is not credited with large end-to-end recall wins.
2. Equivalence-class-aware candidate compression. We raise Fine synonym crowding from a post-processing detail to a stated research problem: a k -item decision budget should be spent on k distinct canonical concepts, so candidates must compete in a quotient space rather than letting one concept’s surface forms occupy several frontier slots. Our current heuristic, `compat_parallel`, is one implementation of that requirement; the requirement is load-bearing, the particular gate predicate is not.

3. State-consistent local-to-global propagation. Local evidence updates change final decisions only if they are written back into the shared belief state and thereby reshape the global candidate pool. Single-champion elimination is rejected as a design default; on our OX slice a forced single-champion cap does not lower end-task F1, so we report that null and retain the argument as an audit concern. Open-XDDx gives the most direct support for write-back and closed-panel decoding.
4. Stage-attributable evaluation. We require a strict failure cascade

TREE_PARENT_ABSENT $\rightarrow$ L2_MISS $\rightarrow$ LOCAL_ELIMINATION $\rightarrow$ GLOBAL_MISRANK $\rightarrow$ M

because our audits show that an unstratified reading reports scoring-interface failures as recall failures—and then motivates interventions that measurably hurt accuracy.

Candidate-relative selectors such as anti-prototype anchoring (anti-anchor) and optional discrimination-compiler hints remain method components and open hypotheses as to strategy; the case-adaptive axis that feeds them is supported by ablation.

Reporting standard: unmodified scoring interface. Name-concept detachment is a real failure mode, and a repair operator for it exists (synonym bind-repair, Section 4.6). We nevertheless report all results with that repair disabled. The reason is methodological rather than technical: applying the same repair to external baselines raises their scores by a comparable or larger margin than ours, so any gap measured with the repair enabled would partly reflect name-normalization coverage rather than hierarchical reasoning. Treating the unmodified interface as the standard keeps the comparison attributable to the reasoning stack, and relegates interface alignment to a protocol issue that any system may adopt.

2 Related Work

Flat LLM diagnosis and RAG. Direct CoT, self-consistency, self-refine, and RAG baselines emit ordered disease lists, optionally conditioned on retrieved passages. They improve access to knowledge but leave candidates as a flat set whose redundancy grows with recall.

Verification and multi-agent collaboration. Dual-Inf uses disease-to-symptom reverse reasoning over a flat differential [11]. MDAgents and MAC structure interaction protocols [1, 9]; MEDDxAgent unifies diagnosis-agent tooling [6]. These methods rarely maintain an explicit shared belief state that local evidence must update before global decoding.

Hierarchies in medicine. KG4Diagnosis routes through specialist layers [2]; H-DDx uses ICD structure mainly for evaluation [4]; DeepRare targets rare-disease tooling [3]. Our hierarchy is a temporary case-specific frame, not a claim of ontology uniqueness.

Interactive versus static diagnosis. SDBench-style settings optimize what to ask next [8]. We study static vignettes where the alignment problem is organizational and state-propagational rather than acquisitional.

3 Problem Formulation

3.1 Task

Input. A static vignette V ; observed facts F traceable to spans in V ; and a shared corpus K available to all retrieval-using methods.

Output. An ordered list of at most k concrete diseases; a case-adaptive diagnostic tree; and an audit of which facts updated which candidates and whether those updates entered the global state.

Objective. Improve end-task ranking under bounded budgets while preserving alignment among (i) candidate strings and canonical concepts, (ii) local evidence and global belief state, and (iii) stage-wise error labels and the interventions they justify. Internal scores are relative ranking scores, not clinically calibrated probabilities.

Out of scope. Active questioning, test selection, treatment, multi-primary comorbidity, pixel-level imaging, and clinical deployment claims.

3.2 From Recall–Discrimination to Alignment Preservation

Let $\mathcal{C}$ be the candidate set after recall. Increasing $|\mathcal{C}|$ can raise coverage, but it also multiplies synonym fragments and mixes abstraction levels, so the classical dilemma is really a statement about how the enlarged set is represented and carried between stages. We make that explicit by naming three invariants that a hierarchical diagnostic pipeline should preserve.

A1 (representation alignment). Two candidate strings denoting the same clinical concept must occupy one competitive slot. Violation: near-synonyms compete as distinct leaves and jointly consume the frontier budget.

A2 (state alignment). Evidence consumed at a local stage must update the belief state that the next stage reads. Violation: a locally revised posterior is discarded when the global candidate pool is formed, so local reasoning is computed and then ignored.

A3 (interface alignment). The emitted name must bind to the canonical concept the scoring contract expects. Violation: a correct leaf is ranked highly, yet the evaluation map records no binding and the case is scored as a miss.

Each invariant maps to one contribution: A1 to equivalence-class compression, A2 to write-back and bounded decoding, A3 to name normalization treated as a protocol obligation. Case-adaptive organization is the substrate that makes all three checkable, because it defines what a family, a slot, and a state are for the current case. Stage-attributable evaluation is the audit that tells us which invariant a given failure violated—without it, an A3 violation is easily reported as insufficient recall.

3.3 Stage Failure Taxonomy

A failed case must be attributed to the earliest applicable stage in the cascade

`TREE_PARENT_ABSENT` $\rightarrow$ `L2_MISS` $\rightarrow$ `LOCAL_ELIMINATION` $\rightarrow$ `GLOBAL_MISRANK` $\rightarrow$ `MAPPE`

and later stages are examined only if all earlier ones succeeded. We use these exact labels throughout, including in released audit records.

`TREE_PARENT_ABSENT`. No clinically acceptable L1 parent exists for the gold (or gold-equivalent) disease; the chosen axis cannot host it. This is a genuine structural miss.

`L2_MISS`. An acceptable parent exists, but no L2 leaf—exact or synonym-equivalent—representing the gold disease exists in the live tree.

`LOCAL_ELIMINATION`. The gold-equivalent leaf exists in the correct family but is dropped by within-family ranking or per-family budget before global decoding.

Table 1: Each stage licenses a different intervention. Misattributing a late stage to an early one motivates the wrong fix, which is how coverage-driven interventions become harmful.

Stage	Invariant	Correct response	Wrong response if misread as recall failure
TREE_PARENT_ABSENT	organization	revise axis; gap-fill	—
L2_MISS	organization	branch-conditioned recall	—
LOCAL_ELIMINATION	A2	bounded frontier; relax family cap	inject more leaves
GLOBAL_MISRANK	A2	posterior write-back; wider pool	inject more leaves
MAPPER_UNBIND	A3	fix name-binding contract	inject leaves / escalate gap-fill

GLOBAL_MISRANK. The leaf reaches the global pool or frontier but is ranked below the protocol’s cutoff.

MAPPER_UNBIND. The leaf is present and ranked high enough, yet the scoring interface records an empty binding (or equivalent name detachment), so the case is counted as a miss.

Why the cascade is a contribution, not logging. Our audits make the risk concrete. Among apparent coverage misses on the DiagnosisArena slice, roughly 18 of 20 were MAPPER_UNBIND and only about 2 were true TREE_PARENT_ABSENT; the intermediate diagnostic PARENT_NOT_IN_L1_SET was empty. An unstratified reading of the same 20 cases says “the hierarchy fails to recall 20% of gold diseases,” which points to leaf injection and gap-fill escalation. When we actually ran that intervention—injecting the full leaf inventory and re-running typed mapping—option @1 fell substantially, because the added candidates perturbed bindings and relative ranks for cases that had been correct. The two remaining absences were axis mismatches that gap-fill does not repair. So the cascade is what separates a fixable interface defect from a structural one, and it is the reason we treat name normalization as a protocol obligation (A3) rather than a recall problem.

3.4 Notation

L1 / L2. Family on a single case-chosen axis / concrete disease leaf under a family.

Canonical concept / quotient space. Equivalence class of synonym or near-synonym disease strings that should share one frontier slot.

Fine crowding. Multiple members of one equivalence class occupy distinct top ranks.

Soft prior. Overturnable L1 relative score used during global arbitration.

Belief state. The shared tree object carrying leaf identities, family membership, and posteriors used by later stages.

4 Method: Alignment-Preserving Hierarchical Hypothesis Management

Figure 1 summarizes APHHM. Modules R1–R5 are the reasoning stack; R3 enforces A1, R4–R5 enforce A2, and A3 is handled as a protocol obligation in Section 4.6 rather than as a scored component.

R1 Organization: multi-source recall $\rightarrow$ case-adaptive L1 (+ optional non-reducing gap-fill) R2 Evidence (open): candidate-relative fact selection $\rightarrow$ sparse L1 updates R3 Compression: equivalence-class merge / exclusive calibrate (compat_parallel heuristic) R4 Propagation: local evidence $\rightarrow$ posterior write-back $\rightarrow$ belief state R5 Decoding: uncertainty-aware bounded frontier or posterior pool $\rightarrow$ global shortlist Scoring interface held fixed for all systems; bind-repair disabled (Section 4.6).

Figure 1: APHHM overview. R1–R5 are the reasoning/alignment modules evaluated in this paper. Interface-level name repair is diagnosed but excluded from reported scores.

4.1 R1: Case-Adaptive Hypothesis Organization

Multi-source concrete-disease recall. Diseases are gathered from case-report corpora, guideline/textbook corpora, and zero-temperature LLM enumeration, then merged by reciprocal rank fusion (RRF) [10].

Single-axis L1 partition. An LLM proposes L1 families under constraints: one shared axis; labels broader than a single disease; near-exclusivity among siblings; a residual family; optional can’t-miss low-prior families. The partition is a temporary comparison coordinate system for the current syndrome, not a universal ontology.

Gap-fill as structural repair, not a proven recall win. If a high-ranked recalled disease has no home, one non-reducing repair call may enlarge or add families. We keep this operator because TREE_PARENT_ABSENT is a real stage that nothing else addresses. We do not claim, on the present locked slices, that enabling gap-fill produces large end-to-end recall or Top- k gains. The audit in Section 3.3 explains why such a claim would be unsafe: most apparent coverage misses were interface unbinds, and the residual absences were axis mismatches that a repair call operating within the existing axis does not fix. Organization is therefore presented as the substrate that makes A1–A3 checkable, not as a recall mechanism with its own headline number.

4.2 R2: Candidate-Relative Evidence Updating (Component; Open Hypothesis)

Salient symptoms often support many families and therefore have low discriminative value. APHHM can select unused facts expected to change relative L1 order.

Anti-prototype anchoring (anti-anchor). Definition. An evidence selector that discounts the most tempting vignette prototype and requires a fact to survive adversary-style comparison across live families (and frozen L2 exemplars when available); it may abstain.

Sparse symmetric update. Each consumed fact updates at most one primary support target and one distinct oppose target among L1 families.

Status in this paper. Anti-anchor and optional discrimination-compiler hints (fact-level direction resources) are retained as implementable components. They are not claimed as the primary cause of the end-to-end improvements reported in Section 6. Pre-registered selector ablations on the DA slice leave option @1 within 4 pp of the deployed configuration (salience selector 0.67 vs. 0.71; no-compiler 0.67 vs. 0.71), both below the 0.10 minimum-explainable threshold and non-significant on paired sign tests (Section 6.2). What is load-bearing is the selector’s input structure: when the L1 axis is not case-adaptive, the same selector abstains and the cascade collapses (Section 6.2). The selector strategy is therefore an open component; the axis that feeds it is not.

4.3 R3: Equivalence-Class-Aware Candidate Compression

This module addresses invariant A1. We state it formally, because we regard it as a research problem rather than a formatting step.

Quotient-space competition. Let L be the live leaf set and let $\sim$ denote clinical equivalence: $\ell_i \sim \ell_j$ when both labels denote the same diagnosis at the granularity the task scores. Write $\pi : L \rightarrow L/\sim$ for the map sending a leaf to its canonical concept. A ranking protocol that reports k items should allocate its budget over classes, not strings: the decision set must satisfy

$$|\pi(\text{Top-}k)| = k,$$

whereas an unconstrained ranker only guarantees $|\text{Top-}k| = k$. The gap between these two conditions is wasted budget. Concretely, if three of the top five leaves belong to one class, the system has spent 60% of its decision budget restating one hypothesis while alternative classes go unexamined.

Why this is not merely cosmetic. Fine crowding degrades accuracy through two distinct channels. First, slot dilution: competing classes are pushed below the cutoff, which manifests as GLOBAL_MISRANK. Second, decision splitting: when the scoring contract expects one canonical name, the ranker’s belief mass is divided across surface forms, so the highest-ranked string may not be the highest-ranked concept. The remedy therefore belongs before scoring rather than inside it: once π is applied at decision time, the interface has no way to recover the intended class. Which of the two channels a given benchmark can expose depends on its scoring contract, and we show in [Section 6.3](#) that closed-set option rematch cannot expose either—so the asymmetric @1/@2 signature on DA, while real as an observation, is not usable as evidence for this module.

Current heuristic: `compat_parallel`. A gold-blind FineCrowdGate inspects whether the top ranks are synonym-crowded. If triggered, only synonym merge is applied (`merge_only`); otherwise only closed-pool Top- K calibration with L1 fallback is applied (`calib_only`). The operators are applied exclusively rather than in series, a design choice we originally justified by operator interference; the ablation withdraws that justification, since serial composition turns out to be indistinguishable from exclusive routing on our slice ([Appendix C](#)). Merge is single-linkage over the synonym relation, so the surviving representatives of any case are pairwise non-synonymous by construction—the delivered set is a genuine quotient, not merely a shorter list. On the MCR slice the gate fires on 82/101 cases and the operator removes 56.5% of surviving candidates ($4.64 \rightarrow 2.02$ per case).

A structural property that constrains what can be measured. Each class is represented by its best-ranked member, and classes are ordered by that member’s rank. Merge is therefore a pure deletion operator: the globally rank-1 leaf heads every partition, so no change of class membership can move rank 1. We verified this invariance on 15,400 perturbed partitions (rank-1 representative unchanged in 100%). This is a property of the deployed rule, not a limitation of our evaluation, and it has a direct consequence we act on in [Section 6.3](#): any endpoint that reads only the top slot is algebraically insensitive to this module and must not be used to test it.

Claim boundary. `compat_parallel` is one implementation of equivalence-aware compression, not a finalized theory of medical synonymy. Our ablations ([Section 6.3](#)) support the quotient requirement while showing that the particular gate predicate, its timing, and serial-versus-exclusive composition are not load-bearing; we therefore state the contribution at the level of the structural requirement and report the predicate as an implementation detail.

4.4 R4: State-Consistent Local-to-Global Propagation

Branch-conditioned L2 generation builds leaves under each live L1 family with per-family budgets and consistency checks. Local evidence annotation then produces case-conditioned leaf support.

Write-back requirement. If those local posteriors are not written into the shared belief state, global decoding still reads “cold” build-time scores. Live posterior write-back copies local posteriors onto leaves so that the global pool reflects the same evidence the local stage just used. A per-family candidate cap is applied in the deployed configuration but is not load-bearing on our OX slice (Section 6.4).

Empirical anchor (OX). On Open-XDDx, a fair closed shortlist over cold trees at the locked budget yields micro-F1 0.576; the same budget with live write-back reaches 0.651 (paired case-bootstrap $\Delta=+0.075$, 95% CI [0.036, 0.113], two-sided add-one $p=4.0\times 10^{-4}$). Changing the evidence budget alone moves F1 by at most +0.008 at either write-back setting. Force-emitting missing leaves did not add leaves on this slice and did not beat the no-emit write-back arm—supporting state consistency over open-set leaf invention as the operative mechanism.

4.5 R5: Bounded Global Decoding

Against the single champion. A natural hierarchical design lets each family submit exactly one winner. We argue this is the wrong default as a design rule, because it contradicts the rest of the pipeline: a system that spends branch-conditioned retrieval to keep a long-tail disease alive should not delete it on the basis of a single within-family comparison made under sparse evidence. If family g submits b_g candidates, single-champion decoding fixes $b_g \equiv 1$ however close the local margin is, so every local ranking error becomes unrecoverable. Worse, the resulting LOCAL_ELIMINATION is invisible downstream, because the global stage only ever sees survivors—a real stage in the audit taxonomy even when end-task F1 does not move.

APHHM instead makes b_g uncertainty-aware:

- a bounded local frontier, with $b_g = 1$ when the local margin is decisive and $b_g = 2$ when the top two are close or the evidence is thin; and/or
- a posterior pool, which drops per-family quotas entirely and admits the Top- N leaves of the belief state for closed decoding.

Single-champion decoding survives only as the special case $b_g = 1$, selected by evidence rather than imposed by construction. On OX, closed live-MAC decoding takes the posterior-pool route: multi-doctor discussion is restricted to the Top- $N=15$ leaves of the written-back state, and outputs are projected back into that pool before emitting $K=5$ names [9]. This is also why A2 and this module are inseparable in the OX result: the pool is only meaningful if the posteriors that define it were updated. Empirical caveat. Forcing $b_g=1$ (cap=1) on the same OX slice does not lower micro-F1 relative to the deployed cap=6 (0.657 vs. 0.651), and removing the cap entirely (cap=999) is likewise within noise (0.649). The single-champion bottleneck is therefore a design and audit concern that this slice does not expose as an end-task cost; we retain the argument and report the null (Section 6.4). L1 scores enter only as soft priors; incomparable within-family scores are not concatenated into a fake global probability.

4.6 Interface Alignment: Diagnosed, Not Credited

Benchmarks differ in how a ranked belief state becomes a score, and that conversion can fail independently of reasoning quality. On DA, a relation-aware answer mapper sometimes writes

an empty option-to-leaf binding even when a synonym-equivalent leaf sits in the ranked shortlist; this is the MAPPER_UNBIND stage of Section 3.3. A repair operator is straightforward: fill empty bindings by synonym/bridge matching against the current shortlist and re-score without re-running the typed mapper.

We nonetheless exclude this operator from every number we report, for two reasons.

First, the lever is not method-specific. Applied offline to the same frozen projections, it raises external baselines by roughly +0.02 to +0.08 option @1—for example, the strongest baseline improves from 0.62 to 0.67, and a self-refine baseline by +0.08—while improving our stack by about +0.02. Because baselines have more unbound gold options to recover, enabling repair for everyone narrows the measured gap and mixes name-normalization coverage into a reasoning comparison. Reporting the unmodified interface is therefore the conservative choice, and it is also the more informative one: it isolates R1–R5.

Second, interface repairs are easy to over-credit. An earlier implementation of ours awarded bridge-match credit from a self-referential retrieval chunk with maximal similarity, which promoted essentially any empty binding to the top option and inflated scores; the affected tables were retracted. This experience is itself evidence for Contribution 4: an interface-stage operator can silently manufacture apparent recall gains unless the failure cascade and the scoring contract are audited separately.

The scientific claim we retain is diagnostic, not competitive: MAPPER_UNBIND dominates apparent coverage misses on DA, so name normalization deserves protocol-level standardization across the field rather than per-paper opt-in gains.

MCR projection. MCR uses open single-trajectory scoring plus LLM reasoning-fact recall over the same hierarchical ranking stack; no option-mapping interface is involved.

5 Experimental Setup

5.1 Datasets and Slices

Frozen 100-case sequential slices: DA d2_seq100_v1 (option-mapped Top-2); MCR mcr_val_seq100_v1 (single-trajectory Acc + LLM reasoning recall); OX ox_seq100_v1 (multi-gold micro-F1 at $K=5$ + LLM interpretation accuracy). Gold never enters runtime generation, evidence selection, write-back, or arbitration. Slices are mechanism-locking cohorts, not full-benchmark leaderboards. All arms are scored with the unmodified evaluation interface (Section 4.6).

5.2 Backbone, Knowledge, Judges

Shared API backbone: meta-llama/llama-3.3-70b-instruct [12]. Shared RAG indices for retrieval arms. MCR/OX judges: Gemini 2.5 Flash with fixed prompts (no lexical substitution for official interpretation / reasoning-recall numbers).

5.3 Baselines

Direct CoT; CoT+RAG; flat retrieve-rerank; flat beam; Dual-Inf; MDAgents; MAC; MED-DxAgent; self-consistency and self-refine; MedPrompt-style and MedRAG-style RAG; iterative i-MedRAG.

Budget-matched flat control. Comparing a multi-stage hierarchical system against a two-call flat baseline is not informative about mechanism. We therefore construct a flat control whose inference budget approaches the hierarchical stack’s, rather than merely exceeding the native baseline. An earlier “matched” variant equalized only a structural proxy and used about 9 model calls per case—an order of magnitude below the hierarchical pipeline’s cached call volume—so it

Table 2: Main results on 100-case slices, all with the evaluation interface unmodified (no bind-repair for any system). Absolute scores are not comparable across datasets.

Dataset	Metric	APHHM configuration	Score	Best external
DA	@1 / @2	organization + equivalence compression	0.71 / 0.78	0.62 / 0.71 (B07)
MCR	Acc / RR	hierarchical ranking stack	0.50 / 0.753	0.24 / 0.412 (B07)
OX	F1 / IAcc	write-back + closed pool	0.651 / 0.355	0.570 / 0.221 (B06)

was not in fact budget matched. The control used here, B02-SC10, runs the flat retrieve-rerank pipeline with ten self-consistent trajectories fused by reciprocal rank, reaching roughly 90 calls per case (92.4 on DA, 89.8 on OX, 93.2 on MCR). This is the strictest flat control available to us and the one we use for the budget argument.

5.4 What Each Result Is Allowed to Support

Contribution 1. Pre-registered axis ablations on DA: a case-irrelevant random partition costs 0.34 option @1 on the full sample (exact sign $p \approx 0$; Holm-surviving confirmatory contrast); the same selector abstains when the axis is not case-adaptive (mean selected facts $4.94 \rightarrow 0.13$). Gap-fill is not credited with large recall wins (Section 6.2).

Contribution 2. Pre-registered ablations on MCR: removing equivalence handling costs 0.102 Acc@1 (Holm-corrected $p = 0.030$); compression removes 56.5% of candidates at no accuracy cost and better graded ranking; a candidate-count-matched semantics-blind merge is worse ($p = 0.015$). The gate predicate is an implementation detail (equivalent within ± 3 pp). The earlier DA claim ($0.59 \rightarrow 0.71$ option @1) is retracted as a closed-set scoring artifact (Section 6.3).

Contribution 3. OX write-back at locked budget: cold $\rightarrow$ hot F1 $0.576 \rightarrow 0.651$ (paired micro-F1 bootstrap $p = 4.0 \times 10^{-4}$); budget alone moves ≤ 0.008 ; closed-panel decode beats posterior Top- K on the same hot tree; per-family cap is not load-bearing (Section 6.4).

Contribution 4. Audit dominance of MAPPER_UNBIND over TREE_PARENT_ABSENT among apparent coverage misses; retraction of an interface-inflated result; full-tree leaf injection with re-typed mapping lowers option @1 from 0.72 to 0.42 (paired 39/9, $p = 1.5 \times 10^{-5}$; Section 6.5).

Not claimed as proven. Anti-anchor / discrimination compiler as the main driver of Table 2 (selector-strategy exclusion holds; axis structure does not).

6 Results

6.1 End-Task Comparison under the Unmodified Interface

Table 2 reports a single cut per dataset. On DA the compressed hierarchical ranking reaches option @1/@2 = 0.71/0.78 (MRR@2 = 0.748); an independently frozen run of the same configuration gives 0.72/0.78, which we treat as the run-to-run range rather than a separate result. On MCR it reaches single-trajectory accuracy 0.50 with LLM reasoning recall 0.753. On OX it reaches micro-P/R/F1 = 0.631/0.672/0.651 with LLM interpretation accuracy 0.355 (357/1007 items). No row includes interface bind-repair for any system, ours or baseline.

Table 3: L1 axis ablations on DA ($n=100$, synonym bind-repair off). Δ is M00-arm. Full-sample is the default reporting caliber; the nonempty row conditions on cases that produced a joint ranking.

Axis	Caliber	@1	@2	$\Delta@1$	disc. b/c
Case-adaptive (deployed)	full	0.71	0.78	—	—
Fixed ICD/specialty	full	0.51	0.60	+0.20	25/5
Random (case-irrelevant)	full	0.37	0.42	+0.34	40/6
Fixed ICD/specialty	nonempty ($n=89$)	0.57	—	+0.135	17/5
Random (case-irrelevant)	nonempty ($n=55$)	0.67	—	+0.018	7/6

6.2 Ablation: Case-Adaptive Organization (Contribution 1)

This subsection asks whether the L1 axis must be decided by the case, or whether any equal-cardinality partition would do. We compare the deployed case-adaptive axis against a frozen ICD/specialty taxonomy and against a case-irrelevant random partition that preserves family count and candidate budget. A flat no-L1 arm was run but is excluded here: its leaf budget collapsed to 28% of the deployed pool and its ranking depth collapsed to 1 by construction, so it cannot support either direction of the claim.

Default caliber: full sample. Table 3 reports option @1/@2 on the DA slice under the unmodified interface. The confirmatory contrast is the random axis: $\Delta@1 = +0.340$ (exact sign 40/6, $p \approx 0$), which survives Holm correction among the five pre-registered confirmatory contrasts (Appendix F). A fixed ICD axis also loses ($\Delta@1 = +0.200$, 25/5), and does so despite emitting more leaves than the deployed configuration (30.1 vs. 17.8 per case)—so the loss is not a coverage deficit.

Cascade: why empty rankings are a consequence, not a bug. The same candidate-relative evidence selector runs under every axis. Mean selected L1 facts fall from 4.94 (deployed) to 0.65 (fixed ICD) to 0.13 (random); empty joint rankings rise from 1 to 11 to 45. Every empty ranking on the random arm has `stop_reason=selector_abstained` with zero facts selected. When families are not case-coherent, there is nothing for a contrastive selector to prefer, so it abstains and the cascade never produces a ranking the mapper can score. We therefore treat empty-ranking rate as a process endpoint of axis quality, and keep the full sample as the default narrative caliber. Conditioning on nonempty rankings is a stricter advanced check: under it the random-axis accuracy residual collapses (7/6), while the fixed-ICD residual remains (17/5, $p=0.017$).

Selector strategy vs. selector input. Replacing the deployed anti-anchor selector with a salience-driven alternative, or removing discrimination-compiler hints while keeping anti-anchor, each costs 0.04 option @1 (11/7 and 9/5; both $p > 0.4$), below the 0.10 minimum-explainable threshold. Pre-registered exclusion therefore holds: end-to-end gains are not explained by the selector strategy. Crossed with the axis result above, the load-bearing object is the selector’s input structure—a case-adaptive partition—not which contrastive rule consumes that partition.

Boundaries. Axis effects are mediated by evidence selection in this design and cannot be separated from it. The slice used here is the DA confirmatory-copy cohort; the plan’s primary registration named a not-yet-materialized development freeze. We claim that the axis must be case-adaptive, not that any particular axis-proposal prompt is optimal.

Table 4: Equivalence handling on MCR, both execution sites crossed. Paired exact tests on the 98 commonly scored cases; Δ is against the deployed configuration. Only the joint removal survives Holm correction across the 15 tests of this family.

Configuration	Sites	Acc@1	any-hit@5	open-MRR	cand./case
Deployed (compat_parallel)	both on	0.50	0.58	0.53	2.02
No decision-time routing	build only	0.50	0.58	0.52	4.64
No build-time de-duplication	decision only	0.50	0.55	0.52	1.81
Neither	none	0.42	0.56	0.48	2.11

6.3 Ablation: Equivalence-Class Compression (Contribution 2)

This subsection reports the pre-registered ablation of the compression module. Because the module is a deletion operator with an invariant top slot (Section 4.3), we first fixed which endpoints are able to respond to it at all, and we report that audit before any comparison.

Endpoint admissibility. For every candidate endpoint we counted, over 200 randomized re-partitions per case, how many cases take more than one value. An endpoint with zero such cases is an algebraic identity: its null variance is exactly 0 and no p -value computed from it carries information. Top-1 endpoints are identities under re-partition on both slices (MCR 0/42, DA 0/35; null s.d. = 0.0000), as the deletion property predicts. On DA, every endpoint we could construct is an identity or near-identity, including after we removed both structural locks (see the dissociation result below). On MCR two endpoints are live: any-hit@5 (gold matched anywhere in the delivered top-5) and open-MRR@5 (reciprocal rank of the first judged-correct delivered label, 0 if none). We therefore judge this module on MCR with those two endpoints plus Acc@1 reported as an equivalence bound. All values use the frozen judge of Section 5.2 at $K=5$ on 98 commonly scored cases. open-MRR@5 is a ranking metric over delivered diagnoses and is unrelated to the reasoning recall (RR) column of Table 2, which measures coverage of reasoning facts; the two are not comparable.

Execution-site factor. Equivalence handling occurs at two places: semantic de-duplication while the hypothesis tree is built, and gated compression at decision time. Table 4 crosses both.

Necessity, and a substitutive relationship between the two sites. Removing equivalence handling at both sites costs 0.102 Acc@1 on the 98 paired cases—roughly a fifth of the deployed configuration’s accuracy—with exact paired $p = 0.002$ (Holm-corrected $p = 0.030$ across the 15 tests of this family; all 10 discordant cases favour the deployed arm) and a 95% interval on the paired difference of $[+3.9, +10.2]$ pp. Removing either site alone leaves Acc@1 at 0.50: the marginal effects sum to 0.031 while the joint effect is 0.102, i.e. strongly super-additive. The two sites are thus largely substitutive—an or-gate—so the architecture is redundant in a way we report as robustness rather than discovering later as dead weight. This also means a single-arm ablation of either site is uninformative by construction, which is why we ran the full cross rather than the one arm our pre-registration originally specified.

Compression is free, and improves graded ranking. Decision-time compression removes 56.5% of surviving candidates ($4.64 \rightarrow 2.02$ per case). Accuracy does not pay for this: with only one discordant case, the paired difference in Acc@1 is bounded to $[-1.0, +1.0]$ pp at 95%. Graded ranking improves: open-MRR $0.52 \rightarrow 0.53$ with a 95% interval of $[+1.3, +7.1]$ pp, excluding zero ($p = 0.016$ unadjusted; an independent operator-level replay of the same contrast reproduces the direction with $[+2.1, +8.2]$ pp, $p = 0.008$). We stress the form of this result: in a paired design the difference is bounded by the discordant count over n , so a small number of moving

Table 5: Predicted dissociation. The closed-set caliber compresses more and responds not at all; its apparent option-@1 gain is the retracted artifact discussed below.

	DA (closed-set rematch)	MCR (open-set)
Candidates per case	4.60 $\rightarrow$ 1.47	4.64 $\rightarrow$ 2.02
Reduction	-68.0%	-56.5%
Open-set @1, Δ	+0.0000	+0.010
Graded ranking, Δ	-0.0028	+0.071, CI [+1.3, +7.1] pp
Under re-partition	identity (s.d. 0.0000)	live ($p = 0.015$)
Closed-set option @1, Δ	+0.112 (retracted)	n/a

cases yields a narrow interval rather than an underpowered test. Reported this way, most arms of this family are equivalence results with explicit bounds, not failures to reject.

Specificity: the semantics matter, not the candidate count. The obvious alternative explanation is that any shortening of the candidate list would help. We reject it three times. First, a candidate-count-matched, semantics-blind merge—same gate, same number of classes, same class-size profile, but randomized membership—delivers an identical mean of 2.405 surviving labels and is worse on both live endpoints: any-hit@5 0.571 versus a 200-draw null of 0.475 ± 0.039 and open-MRR 0.486 versus 0.444 ± 0.014 , both $p = 0.015$. Second, accuracy is not monotone in candidate count across Table 4: $1.81 \rightarrow 0.50$, $2.02 \rightarrow 0.50$, $2.11 \rightarrow 0.42$, $4.64 \rightarrow 0.50$. Third, the cross-caliber dissociation below produces a larger candidate reduction with exactly zero effect. The mechanism is visible in the failure mode: our merge absorbs a leaf only into a synonymous representative, so a gold match is inherited by the representative, whereas blind merge absorbs leaves into unrelated representatives and destroys the match. Semantics-blind merge also leaves 0.79 synonymous duplicate pairs per draw among survivors, against 0.00 for the deployed operator in 100% of cases.

Cross-caliber dissociation, and a retraction. The module acts on the candidate space the system itself constructs, so its value should scale with how much of that space the system must disambiguate. Closed-set option rematch supplies a pre-enumerated, mutually exclusive answer set and credits absorbed leaves through the member-to-representative map, i.e. the scoring contract performs the de-duplication on the system’s behalf. The architecture therefore predicts an effect under open-set scoring and none under closed-set rematch. Table 5 confirms both halves, and the null half is structural rather than empirical.

An earlier version of this work reported the last row as the primary evidence for this contribution: raw joint ranking at option @1 ≈ 0.59 against 0.71 with compression. We audited that gain and now retract it. The two arms emit character-identical rank-1 labels in all 89 gated cases, so no reordering occurs; in the 12 cases where a gold-bound option is promoted to rank 1, the delivered rank-1 label already matched gold under open scoring in 12/12; promotion is not selective for gold (Fisher $p = 0.199$); it is unidirectional because absorption can only raise an option’s binding, converting a strict loss into a tie that dense ranking credits as top-1; and 0/12 of the promotions join concept-identical labels—all are coarsenings such as microvenular hemangioma into hemangioma. The effect is exactly zero once member-to-representative credit is removed ($\Delta = +0.0000$ on open @1). We report this retraction because it is the sharpest available test of our own evidence: the closed-set caliber manufactures a favourable number that the open-set caliber does not, and the MCR results above survive a filter the DA result failed.

The requirement is load-bearing; the heuristic is not. Pre-registered variants of the operator—unconditional merge, calibration only, serial rather than exclusive composition, frequency-

Table 6: OX micro-F1 under a 2×2 of evidence budget and write-back (cap=6, closed live-MAC decode, $n=100$). Write-back moves ≈ 0.075 ; budget moves ≈ 0.008 .

	Cold (no write-back)	Hot (write-back)
Locked budget (L1=4)	0.576	0.651
Default budget (L1=6)	0.584	0.658

matched random routing, and a stricter concept-identity predicate—are all equivalent to the deployed gate within ± 3 pp on the audited endpoints (Table 11). Two of them are structural identities in this setup and produce no difference at any n . We therefore state the contribution at the level of the quotient requirement and demote the gate predicate to an implementation detail, as our pre-registration specified should happen if the concept-identity variant reached parity. A structural claim that survives changes of its own implementation is stronger evidence for the requirement than one tied to a particular threshold.

Limitations. Of the 15 paired tests in this family only the joint removal survives Holm correction; the remaining contrasts are reported as bounded equivalences, not as effects. The execution-site cross exists on MCR only, and we do not claim cross-caliber replication of it—the dissociation above is why we regard the closed-set slice as a predicted boundary rather than a missing replication, but that argument is theoretical and the DA cross was not run. Both live endpoints were selected after Top-1 endpoints were found to be identities; the selection criterion is the structural audit, which is independent of any p -value, but the endpoints still require confirmation on an independent slice. Sign tests on this slice need six unidirectional moving cases for $p < 0.05$, which the smaller contrasts do not reach at $n \approx 100$.

6.4 Ablation: State Consistency and Bounded Decoding (Contribution 3)

This subsection decouples three operators that the OX main result bundles: evidence budget, posterior write-back, and closed-panel decoding. Table 6 crosses budget with write-back at a fixed per-family cap of 6.

Write-back is load-bearing; budget is not. At the locked budget the confirmatory cold arm loses 0.075 micro-F1 against the deployed configuration. A paired case-bootstrap of micro-F1 (the registered OX diagnostic aggregate) gives $\Delta = +0.075$ with 95% CI [0.036, 0.113] and two-sided add-one $p = 4.0 \times 10^{-4}$ (5000 resamples); the contrast survives Holm correction among the five confirmatory tests (Appendix F). An auxiliary per-case F1 sign test agrees in direction (mean $\Delta = +0.073$; 42/14; $p = 2.3 \times 10^{-4}$). The clean budget contrast—default vs. locked, holding write-back fixed—moves F1 by at most +0.008 on either side of the write-back factor, with interaction -0.001 . An earlier draft reading that attributed the gain to budget calibration compared the joint (budget+write-back) arm against cold and is retracted: that contrast confounds the two factors. Pre-registered falsification clauses “write-back is inert” and “budget alone explains the gain” are both unmet.

Closed-panel decode beats posterior Top- K on the same hot tree. Holding the written-back tree fixed and changing only the decoder: closed live-MAC micro-F1 0.651, posterior-pool MCR 0.610, global posterior Top-5 0.593. Paired case-F1 favours the closed panel against posterior Top-5 (mean $\Delta = +0.056$, 32/9, $p = 4 \times 10^{-4}$) and against posterior-pool MCR (mean $\Delta = +0.040$, 30/13, $p = 0.014$). Interpretation accuracy stays within 0.35–0.36 across the three, so the gain is diagnostic rather than explanatory. This isolates bounded closed decoding as an increment on top of write-back, not a substitute for it.

Table 7: Flat retrieve-rerank under increasing inference budget, unmodified interface. B02-SC10 is the budget-matched control; the earlier structural-proxy variant was not truly matched.

Dataset	Metric	native (≈ 2 calls)	proxy (≈ 9 calls)	SC10 (≈ 90 calls)
DA	option @1	0.56	0.48	0.47
OX	micro-F1	0.495	0.479	0.487
MCR	Acc	0.17	0.17	0.15

Per-family cap is not load-bearing on this slice. Forcing a single champion per family (cap=1) yields micro-F1 0.657; removing the cap (cap=999) yields 0.649. Both sit within noise of the deployed 0.651. The pre-registered clause “if uncapping does not harm, delete cap from the mechanism claim” therefore fires: we retain write-back and closed-panel decoding, and demote the cap to an implementation choice. The design argument against imposing $b_g \equiv 1$ by construction (Section 4.5) stands as an audit concern (LOCAL_ELIMINATION remains a real stage); this slice simply does not price that stage in end-task F1.

Emit vs. write-back. The no-emit write-back arm still beats an emit-enabled arm (0.651 vs. 0.645) that emitted zero new leaves—evidence that updating the belief state, not inventing leaves, drives the gain.

6.5 Stage Attribution (Contribution 4)

Table 1 and Section 3.3 give the cascade; here we record what it changed in practice. Stratifying the 20 apparent coverage misses on DA moved roughly 18 of them from “recall failure” to MAPPER_UNBIND, leaving about 2 genuine TREE_PARENT_ABSENT cases. The revised priority is interface standardization and equivalence compression first, indiscriminate leaf injection last. The counterfactual is measurable: full-tree leaf injection with re-run typed mapping—the intervention that unstratified coverage numbers recommend—lowered option @1 from 0.72 to 0.42 ($\Delta = -0.30$; mean +16.1 extra leaves per case; coverage 0.93). On the 100 paired cases the discordance is 39 favouring the uninjected arm against 9 favouring injection (exact sign $p = 1.5 \times 10^{-5}$); that contrast is the confirmatory arm for this contribution and survives Holm correction (Appendix F). It is a historical reuse and has not been re-verified under the frozen judge contract of this draft. We therefore report stage counts alongside end-task scores, and treat any coverage statistic that has not been stratified as uninterpretable.

6.6 Budget-Matched Flat Control (B02-SC10)

Table 7 shows that scaling the flat pipeline to roughly 90 model calls per case does not recover hierarchical performance. Across all three datasets, ten-sample self-consistency with rank fusion stays at or below the native two-call configuration: DA option @1 0.47 versus 0.56 native, MCR accuracy 0.15 versus 0.17, and OX micro-F1 0.487 versus 0.495. Every value remains far below the corresponding APHHM row in Table 2.

Two observations matter for interpretation. First, the earlier proxy-matched variant was never a valid budget control: at about 9 calls per case it sat an order of magnitude below the hierarchical stack, so its weakness could have been attributed to insufficient compute. B02-SC10 removes that objection, and the conclusion strengthens rather than weakens. Second, the failure mode of self-consistency here is informative: on OX, aggregating ten trajectories by rank fusion raised diagnostic micro-F1 slightly while collapsing interpretation accuracy from 0.445 to 0.044, because fusion keeps names but discards the per-trajectory explanation structure. Additional sampling in a flat space buys marginal list reordering; it does not buy an organized hypothesis space or a shared belief state.

7 Discussion

What the hierarchy is for. APHHM does not imitate hospital org charts. It provides a case-specific coordinate system and a shared belief state that later stages must read.

What we no longer over-claim. Gap-fill is not presented as a major recall success. Anti-anchor is not presented as the proven headline mechanism: selector-strategy ablations are directional nulls, while the selector’s input (a case-adaptive axis) is load-bearing (Section 6.2). Interface bind-repair is not presented at all as a result: it is diagnosed as a shared protocol gap, and excluded because it benefits baselines at least as much as our stack. The DA option-@1 gain we previously attributed to equivalence compression is retracted (Section 6.3); the contribution now rests on open-set MCR endpoints, and the gate predicate is reported as an implementation detail rather than as a mechanism. Per-family candidate caps are likewise demoted: uncapping and single-champion forcing are both within noise of the deployed OX configuration (Section 6.4).

On interface levers as a community issue. That exclusion has a wider implication. If a single offline name-matching pass can move published option accuracy by several points for most systems, then differences between recent diagnostic agents may be partly differences in name normalization coverage. We suggest that benchmarks fix the name-binding contract, and that papers report which contract they used, rather than each system supplying its own repair.

Open risks. The quotient requirement is supported on one slice by one Holm-surviving contrast; the predicate-level questions are bounded rather than resolved, and resolving them needs roughly twice our sample or a slice stratified to a balanced gate firing rate (Appendix C). Endpoint admissibility for compression was established structurally but after the fact, and needs independent-slice confirmation. The single-champion bottleneck was not priced by OX end-task F1 under a posterior-pool decoder; stress-testing it needs a route where LOCAL_ELIMINATION can reach the scored shortlist. Axis effects are mediated by evidence selection and were measured on a DA confirmatory-copy slice rather than the not-yet-materialized development freeze. 100-case slices are not full-benchmark significance tests, and B02-SC10 matches call counts rather than audited token ledgers.

A note on reporting nulls. Much of Section 6.3 consists of arms that did not move the endpoint. We report these as intervals rather than as p -values because the paired design bounds the difference by the discordant count over n , which makes such results statements about how small an effect can be rather than failures to detect one. We recommend this form generally for module ablations of pipelines whose operators are, like ours, partly redundant by design: a difference test on a substitutive component is close to uninterpretable, whereas a bound is not.

Safety. Offline research only; not a clinical decision-support claim.

8 Conclusion

We upgraded the Recall–Discrimination Dilemma into alignment-preserving hierarchical hypothesis management. The central problems are not only whether the correct disease is retrieved, but whether candidates compete as canonical concepts, whether local evidence updates the global belief state, and whether evaluation errors are attributed to the correct stage. Under that framing, case-adaptive organization, equivalence-aware compression, state-consistent write-back with bounded decoding, and stage-attributable evaluation are first-class contributions; candidate-relative selector strategies remain open components whose input structure, not

whose particular rule, is what the ablations support. All comparisons are made with the evaluation interface untouched, and against a flat control scaled to comparable inference budget, so that the remaining gaps are attributable to how hypotheses are organized and propagated rather than to name matching or sampling volume. We hope future diagnostic systems fix the name-binding contract before reporting gains, and refuse to treat MAPPER_UNBIND as L2_MISS.

A Glossary of Terms and Repository Codes

APHHM. Alignment-Preserving Hierarchical Hypothesis Management; the framing of this paper.

Recall-Discrimination Dilemma. Increasing recall admits correct diseases but also redundant/similar candidates that harden discrimination; retained as motivation, upgraded by alignment requirements.

L1 / L2. Level-1 family / Level-2 concrete disease leaf.

RRF. Reciprocal Rank Fusion [10].

Anti-anchor. Candidate-relative evidence selector that resists headline prototype bias; open hypothesis in this draft.

Soft prior. Overturnable L1 score used in global arbitration.

Belief state. Shared tree carrying identities, membership, and posteriors for later stages.

A1 / A2 / A3. Representation, state, and interface alignment invariants (Section 3.2).

Fine crowding. Near-synonym leaves occupy multiple top frontier slots, so a k -slot budget covers fewer than k canonical concepts.

Quotient / canonical concept space. The set $L/\sim$ of clinical equivalence classes; competition should be scored there rather than over raw label strings.

open-MRR@ k . Reciprocal rank of the first delivered diagnosis judged correct among the top k , 0 if none. A ranking metric over delivered diagnoses; distinct from reasoning recall (RR), which measures coverage of reasoning facts. The two are not comparable.

any-hit@ k . Whether any of the top- k delivered diagnoses is judged correct.

Count-matched semantics-blind merge. Control operator with the deployed gate, class count and class-size profile, but randomized class membership; isolates equivalence semantics from candidate-count reduction (Section 6.3).

Endpoint admissibility / algebraic identity. An endpoint is inadmissible for a module if its per-case value is constant under randomized perturbation of that module, i.e. null variance exactly 0. Such endpoints yield p -values that carry no information.

Selector abstention cascade. Non-case-adaptive L1 axis $\rightarrow$ candidate-relative selector abstains ($n_{\text{sel}}=0$) $\rightarrow$ empty ranking $\rightarrow$ mapper miss (Section 6.2).

structural_reach. Gold present anywhere in the tree before ranking; a coverage process metric, not end-task @1.

Factorial decomposition. Crossing factors (e.g. budget $\times$ write-back) so main effects are not read off a confounded joint contrast (Section 6.4).

Table 8: DiagnosisArena d2_seq100: option @1/@2 (selected arms), unmodified interface.

Arm	@1	@2	MRR@2
APHHM (organization + compression)	0.71	0.78	0.748
B07 MEDDxAgent	0.62	0.71	0.665
B06 MAC	0.61	0.67	0.640
B04 Dual-Inf	0.60	0.70	0.650
B17 i-MedRAG	0.60	0.67	0.635
B05 MDAgents	0.58	0.67	0.625
B13 self-refine	0.57	0.62	0.595
B02 flat rerank (native)	0.56	0.63	0.595
B00 Direct CoT	0.54	0.61	0.575
B03 flat beam	0.52	0.60	0.560
B02-SC10 (budget matched)	0.47	0.59	0.530

Gap-fill. One-shot non-reducing L1 repair when a high-ranked recalled disease has no parent; not claimed as a major end-to-end recall win here.

compat_parallel. Heuristic exclusive merge-vs-calibrate compressor behind FineCrowdGate.

FineCrowdGate. Gold-blind detector of synonym crowding among top-ranked leaves.

Live posterior write-back. Writing local evidence posteriors into the shared tree before global decoding.

Bounded frontier / posterior pool. Uncertainty-aware multi-leaf submission or Top- N closed decoding set.

closed_live_mac_supervisor. Closed-pool multi-doctor + supervisor shortlist used on OX.

B02-SC10. Flat retrieve-rerank control with ten self-consistent trajectories fused by reciprocal rank (≈ 90 model calls per case); the budget-matched baseline used in Section 6.6.

Synonym bind-repair / Approach A. Offline repair of empty option-leaf bindings at the scoring interface. Diagnosed in Section 4.6 but excluded from all reported results, because it raises baseline scores at least as much as ours.

TREE_PARENT_ABSENT. No acceptable L1 parent for the gold disease.

L2_MISS. Parent exists but no gold-equivalent L2 leaf in the tree universe.

LOCAL_ELIMINATION. Leaf exists in-family but is dropped before global decoding.

GLOBAL_MISRANK. Leaf enters the global pool but ranks too low.

MAPPER_UNBIND. Leaf present/rank-sufficient, but evaluation binding is empty.

F2 / F4 / F6. Evidence prefix lengths in logs; implementation detail only.

Config A. Branch-conditioned L2 generation recipe; implementation detail.

Interp. Acc / IAcc. OX interpretation accuracy under LLM judge template ox.interpretation_consistency.

B Extended Baseline Tables

Values match repository summaries under runs/paper_v1/. Every row uses the unmodified evaluation interface; no arm receives synonym bind-repair.

Table 9: MedCaseReasoning mcr_val_seq100: single-trajectory accuracy and LLM reasoning recall.

Arm	Acc	Reasoning Recall
APHHM hierarchical stack	0.50	0.753
B07 MEDDxAgent	0.24	0.412
B06 MAC	0.23	0.527
B01 CoT+RAG	0.22	0.478
B00 Direct CoT	0.18	0.510
B02 flat rerank (native)	0.17	0.404
B02-SC10 (budget matched)	0.15	—

Table 10: Open-XDDx ox_seq100 ($K=5$). SC10 interpretation accuracy collapses because rank fusion discards per-trajectory explanations.

Arm	micro-F1	Interp. Acc
APHHM write-back + closed pool	0.651	0.355
Cold tree, locked budget (AB13)	0.576	0.352
Cold tree, default budget (AB16)	0.584	0.355
B06 MAC	0.570	0.221
B00 Direct CoT	0.543	0.233
B05 MDAgents	0.543	0.424
B02 flat rerank (native)	0.495	0.419
B07 MEDDxAgent	0.491	0.403
B02-SC10 (budget matched)	0.487	0.044

C Compression Variants: Equivalence Bounds

This appendix reports every pre-registered variant of the equivalence-compression operator (Section 6.3), including those that produce no difference. We report them as equivalences with bounds rather than as null results, for a reason specific to the paired design: the difference between two arms is bounded by m/n , where m is the number of cases discordant on the endpoint, so few moving cases give a narrow interval rather than a weak test. Intervals are exact conditional McNemar intervals (Clopper–Pearson on the discordant pairs, mapped to the difference scale) on $n = 98$ commonly scored MCR cases with the frozen judge at $K=5$. The pre-declared non-inferiority margin is ± 5 pp, half the effect size our pre-registration treats as substantive.

Why any-hit@5 is an identity here. The deployed configuration delivers 2.20 candidates per case on average and fewer than $K=5$ in 86 of 98 cases, so there is no padding headroom: any-hit@5 reduces to whether gold is in the surviving set. Because merge is a deletion operator that always retains the best-ranked member of a class, none of the arms above can drop a label that matched gold, and seven of the nine have zero moving cases on this endpoint. The identical column values across those arms are therefore the same number, not a converging null, and no sample size changes this. We flag it explicitly rather than tabulating it as evidence.

Structural identities. Serial composition (AB09) delivers the identical set, the identical rank-1 leaf, and zero moving cases on all three endpoints in this configuration; removing the L1 soft prior (AB20) moves one case. Neither can produce a difference at any sample size on this slice, so we do not report them as evidence about the mechanism in either direction. This also corrects an earlier reading of AB09 as evidence of operator interference: on the reconstructed pre-compat input the serial arm is not merely similar to exclusive routing, it is indistinguishable from it.

Conditional effect sizes. Restricted to cases where the delivered list actually differs, the deployed arm gains +0.125 Acc@1 over unconditional merge (16 cases) and +0.107 over frequency-matched random routing (28 cases). Conditioning corrects the estimand but does not add

Table 11: Pre-registered compression variants against the deployed configuration. “Output disc.” counts cases whose delivered list differs at all; endpoint columns give the 95% interval on the paired difference in percentage points, positive meaning the deployed arm scores higher. any-hit@5 is an algebraic identity for most arms (see note below) and is reported for completeness only.

Code	Variant	Disc.	Acc@1	open-MRR	Verdict
AB05	no decision-time routing	94	$[-1.4, +2.0]$	$[+2.1, +8.2]$	effect, $p = 0.008$
AB07	unconditional merge	16	$[-1.4, +2.0]$	$[-1.4, +2.0]$	equivalent
AB08	calibration only	82	$[-2.5, +3.0]$	$[-3.4, +5.6]$	equiv., bidirectional
AB09	serial merge $\rightarrow$ calibrate	7	$[0, 0]$	$[0, 0]$	identity
AB10	freq.-matched random routing	28	$[-1.3, +3.1]$	$[-1.3, +3.1]$	equivalent
AB10b	count-matched blind merge	33	$[0, 0]$	$[-1.3, +3.1]$	identity at @1
AB10c	blind merge, rank-1 size matched	26	$[0, 0]$	$[-1.4, +2.0]$	identity at @1
AB11	concept-identity predicate	37	$[-1.4, +2.0]$	$[-1.3, +3.1]$	equivalent
AB20	no L1 soft prior	10	$[-1.0, +1.0]$	$[-1.0, +1.0]$	near-identity

AB10b and AB10c are identities at @1 by construction and are judged instead by the permutation test of [Section 6.3](#), where they are significantly worse than the deployed operator on both live endpoints ($p = 0.015$). AB08 is the one variant whose discordant cases run in both directions (4 for the deployed arm, 2 against), so no unidirectional claim is available for it at any sample size.

Table 12: Selector cascade under three L1 axes on DA ($n=100$).

Axis	mean n_{sel}	$n_{\text{sel}}=0$	empty rank	@1
Case-adaptive	4.94	17	1	0.71
Fixed ICD	0.65	89	11	0.51
Random	0.13	97	45	0.37

power—the sign test still counts only endpoint-moving cases—so we report these only alongside the full-sample intervals above.

Attainable resolution on this slice. The gate fires on 82/101 cases, so any frequency-matched routing contrast can differ from the deployed arm on at most $2 \min(82, 19) = 38$ cases; AB10 already reaches 28 of them. A sign test here needs six unidirectional moving cases for $p < 0.05$, which linear extrapolation puts at $n \approx 196$ for AB10, AB10b and AB11, $n \approx 294$ for AB07 and AB10c, and $n \approx 588$ for AB20. Resolving predicate-level questions therefore requires either a larger slice or a slice stratified to a gate firing rate near 0.5, where the discordance ceiling is maximal. We state this as the cost of the next experiment rather than as a property of the mechanism.

D Axis Cascade and Organization Details

[Table 12](#) expands the process metrics behind [Section 6.2](#). Empty rankings are uniformly selector_abstained with zero selected facts on the random axis; they are the terminal state of the cascade, not mapper failures.

E State-Consistency Factorial Details

[Table 13](#) records the nested cap arms and the decode-only contrast of [Section 6.4](#).

F Five Confirmatory Contrasts (Holm)

[Table 14](#) reports the five pre-registered confirmatory contrasts under Holm step-down at $\alpha=0.05$. Endpoint heterogeneity across blocks is intentional and registered; Holm is applied to the five

Table 13: Nested OX arms at locked budget with write-back, plus decode-only variants on the same hot tree.

Arm	Change	micro-F1	Δ vs. deployed
Deployed	L1=4, wb, cap=6, closed	0.651	—
AB17	cap=1 (single champion)	0.657	−0.006
AB19	cap=999 (uncapped)	0.649	+0.002
AB15-post	decode = posterior Top-5	0.593	+0.058
AB15-pool	decode = post_n_mcr	0.610	+0.041

Table 14: Pre-registered confirmatory family. Effects are oriented so that positive favours the deployed configuration (or, for C5, the magnitude of the selector drop).

ID	Contrast	Endpoint	Effect	p	p_{Holm}
C1	vs. random axis	DA @1 (full)	+0.34	≈ 0	≈ 0
C2	vs. blind merge	MCR any-hit@5	+0.097	0.015	0.030
C3	vs. cold write-back	OX micro-F1 boot.	+0.075	4.0×10^{-4}	1.2×10^{-3}
C4	vs. leaf inject	DA @1 (39/9)	+0.30	1.5×10^{-5}	6.0×10^{-5}
C5	vs. salience selector	DA @1	+0.04	0.48	0.48

confirmatory p -values, not to a common endpoint. C3 uses a paired case-bootstrap of OX micro-F1; C4 uses the exact sign test on per-case `compat_opt1` vs. `typed_opt1`. C5 is an exclusion control whose surviving null is the intended outcome.

G Ethical Considerations

Public research vignettes only. No patient-care decisions. LLM judges may inherit model biases; prompts and judge identities are frozen when reporting official-style numbers.

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